

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

MI-8003(2) (CBGS)

B.E. VIII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Geostatistics

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Explain the following term:
 - i) Standard Deviation
 - ii) Median
 - iii) Skewness
- b) What is Lognormal Distribution? Write down the characteristics of Lognormal Distribution.
2. a) What is ore reserve estimation? Define the term tonnage and tonnage factor?
- b) What is triangular method of reserve estimation? How it is carried out?
3. a) Explain constant distance and inverse distance weighting technique of grade estimation.
- b) Write a note on theory of regionalised variable. Write down the stationary assumptions in geostatistics.

MI-8003 (2) (CBGS)

PTO

[2]

4.
 - a) What is mean by geostatistical prediction? How it is different from simple prediction? Where it can be useful?
 - b) Apart from having sufficient data pairs, identify and describe the fundamental characteristics that a variogram should have before being used for kriging or simulation.
5.
 - a) Derive the relationship between semivariogram, covariogram and variance.
 - b) Draw and explain the following semivariogram models:
 - i) Exponential models
 - ii) Gaussian model
6.
 - a) What is mean by Extension variance and estimation variance? Derive the formula for calculation of Estimation variance.
 - b) What is Kriging? Write down the Kriging system of equations and the formula to calculate Kriging variance.
7.
 - a) Compare Proved reserves with probable reserves.
 - b) What is bench compositing? How it is carried out?
8. How grade tonnage relationship is developed for a mineralized inventory? Explain by giving example.
